

```

File Edit Tabs Help
config.status: executing depfiles commands
config.status: executing libtool commands
pi@pi:~/deepdream/protobuf-2.6.1 $
pi@pi:~/deepdream/protobuf-2.6.1 $
pi@pi:~/deepdream/protobuf-2.6.1 $
pi@pi:~/deepdream/protobuf-2.6.1 $ sudo make install
Making install in .
make[1]: Entering directory '/home/pi/deepdream/protobuf-2.6.1'
make[2]: Entering directory '/home/pi/deepdream/protobuf-2.6.1'
make[2]: Nothing to be done for 'install-exec-am'.
/bin/mkdir -p '/usr/local/lib/pkgconfig'
/usr/bin/install -c -m 644 protobuf.pc protobuf-lite.pc '/usr/local/lib/pkgconfig'
make[2]: Leaving directory '/home/pi/deepdream/protobuf-2.6.1'
make[1]: Leaving directory '/home/pi/deepdream/protobuf-2.6.1'
Making install in src
make[1]: Entering directory '/home/pi/deepdream/protobuf-2.6.1/src'
dephbase= echo google/protobuf/compiler/main.o | sed 's|[/\]"$|.deps/&;|s|\.o$||';\
g++ -DHAVE_CONFIG_H -I. -I... -pthread -Wall -Wwrite-strings -Woverloaded-virtual -Wno-sign-compare -O2 -g -DNDEBUG -MT google/protobuf/compiler/main.o -MD -MP -MF $dephbase.Tpo -c -o google/protobuf/compiler/main.o google/protobuf/compiler/main.cc &&\
mv -f $dephbase.Tpo $dephbase.Po
dephbase= echo google/protobuf/stubs/atomicops_internals_x86_gcc.lo | sed 's|[/\]"$|.deps/&;|s|\.lo$||';\
/bin/bash ../libtool --tag=CCX --mode=compile g++ -DHAVE_CONFIG_H -I. -I... -pthread -Wall -Wwrite-strings -Woverloaded-virtual -Wno-sign-compare -O2 -g -DNDEBUG -MT google/protobuf/stubs/atomicops_internals_x86_gcc.lo -MD -MP -MF $dephbase.Tpo -c -o google/protobuf/stubs/atomicops_internals_x86_gcc.lo google/protobuf/stubs/atomicops_internals_x86_gcc.cc &&\
mv -f $dephbase.Tpo $dephbase.Plo
libtool: compile: g++ -DHAVE_CONFIG_H -I. -I... -pthread -Wall -Wwrite-strings -Woverloaded-virtual -Wno-sign-compare -O2 -g -DNDEBUG -MT google/protobuf/stubs/atomicops_internals_x86_gcc.lo -MD -MP -MF google/protobuf/stubs/.deps/atomicops_internals_x86_gcc.Tpo -c google/protobuf/stubs/atomicops_internals_x86_gcc.cc -fPIC -DPIC -o google/protobuf/stubs/.libs/atomicops_internals_x86_gcc.o
libtool: compile: g++ -DHAVE_CONFIG_H -I. -I... -pthread -Wall -Wwrite-strings -Woverloaded-virtual -Wno-sign-compare -O2 -g -DNDEBUG -MT google/protobuf/stubs/atomicops_internals_x86_gcc.lo -MD -MP -MF google/protobuf/stubs/.deps/atomicops_internals_x86_gcc.Tpo -c google/protobuf/stubs/atomicops_internals_x86_gcc.cc -o google/protobuf/stubs/atomicops_internals_x86_gcc.o >/dev/null 2>&1
dephbase= echo google/protobuf/stubs/atomicops_internals_x86_msvc.lo | sed 's|[/\]"$|.deps/&;|s|\.lo$||';\
/bin/bash ../libtool --tag=CCX --mode=compile g++ -DHAVE_CONFIG_H -I. -I... -pthread -Wall -Wwrite-strings -Woverloaded-virtual -Wno-sign-compare -O2 -g -DNDEBUG -MT google/protobuf/stubs/atomicops_internals_x86_msvc.lo -MD -MP -MF $dephbase.Tpo -c -o google/protobuf/stubs/atomicops_internals_x86_msvc.lo google/protobuf/stubs/atomicops_internals_x86_msvc.cc &&\
mv -f $dephbase.Tpo $dephbase.Plo
libtool: compile: g++ -DHAVE_CONFIG_H -I. -I... -pthread -Wall -Wwrite-strings -Woverloaded-virtual -Wno-sign-compare -O2 -g -DNDEBUG -MT google/protobuf/stubs/atomicops_internals_x86_msvc.lo -MD -MP -MF google/protobuf/stubs/.deps/atomicops_internals_x86_msvc.Tpo -c google/protobuf/stubs/atomicops_internals_x86_msvc.cc -fPIC -DPIC -o google/protobuf/stubs/.libs/atomicops_internals_x86_msvc.o
libtool: compile: g++ -DHAVE_CONFIG_H -I. -I... -pthread -Wall -Wwrite-strings -Woverloaded-virtual -Wno-sign-compare -O2 -g -DNDEBUG -MT google/protobuf/stubs/atomicops_internals_x86_msvc.lo -MD -MP -MF google/protobuf/stubs/.deps/atomicops_internals_x86_msvc.Tpo -c google/protobuf/stubs/atomicops_internals_x86_msvc.cc -o google/protobuf/stubs/atomicops_internals_x86_msvc.o >/dev/null 2>&1

```

Fig. 8: Installing process

```

pi@pi:~/deepdream/protobuf-2.6.1/python $ python setup.py build
Installed /home/pi/deepdream/protobuf-2.6.1/python/.eggs/google_apputils-0.4.2-py2.7.egg
Searching for python-gflags>=1.4
Reading https://pypi.org/simple/python-gflags/
Downloading https://files.pythonhosted.org/packages/df/ec/e31302d355bc9d207d9b858adc1ecc4a6d8c855730c8ba4ddb3d3f8eb8d/python-gflags-3.1.2.tar.gz#sha256=49a131e099ef689e14aa53ca063839c34f6a168afe622217b5b875492a1ee2
Best match: python-gflags 3.1.2
Processing python-gflags-3.1.2.tar.gz
Writing /tmp/easy_install-MVP041/python-gflags-3.1.2/setup.cfg
Running python-gflags-3.1.2/setup.py -q bdist_egg --dist-dir /tmp/easy_install-MVP041/python-gflags-3.1.2/egg-dist-tmp-xbgdpb
warning: no files found matching 'README'
zip_safe flag not set; analyzing archive contents...
Moving python_gflags-3.1.2-py2.7.egg to /home/pi/deepdream/protobuf-2.6.1/python/.eggs
Installed /home/pi/deepdream/protobuf-2.6.1/python/.eggs/python_gflags-3.1.2-py2.7.egg
running build
running build_py
Generating google/protobuf/descriptor_pb2.py...
Generating google/protobuf/compiler/plugin_pb2.py...
Generating google/protobuf/unit_test_pb2.py...
Generating google/protobuf/unit_test_custom_options_pb2.py...
Generating google/protobuf/unit_test_import_pb2.py...
Generating google/protobuf/unit_test_import_public_pb2.py...
Generating google/protobuf/unit_test_mset_pb2.py...
Generating google/protobuf/unit_test_no_generic_services_pb2.py...
Generating google/protobuf/internal/descriptor_pool_test1_pb2.py...
Generating google/protobuf/internal/descriptor_pool_test2_pb2.py...
Generating google/protobuf/internal/test_bad_identifiers_pb2.py...
Generating google/protobuf/internal/missing_enum_values_pb2.py...
Generating google/protobuf/internal/more_extensions_pb2.py...
Generating google/protobuf/internal/more_extensions_dynamic_pb2.py...
Generating google/protobuf/internal/more_messages_pb2.py...
Generating google/protobuf/internal/factory_test1_pb2.py...
Generating google/protobuf/internal/factory_test2_pb2.py...
Generating google/protobuf/pyext/python_pb2.py...
creating build
creating build/lib.linux-armv7l-2.7
creating build/lib.linux-armv7l-2.7/google
creating build/lib.linux-armv7l-2.7/google/protobuf
creating build/lib.linux-armv7l-2.7/google/protobuf/internal
copying google/protobuf/internal/_init_.py -> build/lib.linux-armv7l-2.7/google/protobuf/internal
copying google/protobuf/internal/api_implementation.py -> build/lib.linux-armv7l-2.7/google/protobuf/internal

```

Fig. 9: Installation in progress

The above helps recognise the person in front of the robot (known or unknown). In this code, we will import three modules: face recognition, cv2, and numpy. We will create

different arrays for recognising faces and names. Make sure to write the image file name of that person for correct face recognition.

In the next part of code, we try

to match the face that has been captured by the camera with the array of known faces. If the face matches, the code will run the espeak synthesizer to speak the person's name